SUPPORTING ADULT GUIDANCE

Naming the sides

Introduction to Trigonometry. For any supporting adult — teacher, practitioner, parent or carer. **You do not need to be a mathematics specialist to support this lesson.**

What this lesson does

The learner learns to name the three sides of a right-angled triangle — **hypotenuse**, **opposite** and **adjacent** — **relative to a chosen angle** θ. The key idea is that the hypotenuse is fixed (opposite the right angle) while the opposite and adjacent are defined by their relationship to θ, so they **swap** when the other acute angle is chosen. Naming is practised in every orientation, so names follow the angle, not the position on the page.

Driving Question: *Which side is which — and why does it depend on the angle you choose?*

The mathematics, in plain terms

Trigonometry needs a shared vocabulary for the sides of a right-angled triangle, and this lesson builds it carefully. The **hypotenuse** is the longest side and is **always opposite the right angle**; it does not depend on which acute angle you look at, so its name never changes. The other two names are defined **relative to the angle** θ you choose: the **opposite** is the side **across from** θ (it does not touch θ), and the **adjacent** is the side **next to** θ that is **not** the hypotenuse. The single most important realisation is that **opposite and adjacent are not fixed to particular sides** — choose the other acute angle and the two swap over, because the side that was across from one angle is next to the other. The hypotenuse alone stays put. This is why the lesson deliberately draws triangles in many orientations: a learner who names sides by their position ('the bottom one', 'the tall one') will come unstuck the moment the triangle is rotated, whereas a learner who names them by their **job relative to the angle** is secure everywhere. The classic error to watch for is confusing the **adjacent** with the **hypotenuse** — both can touch θ, but only the hypotenuse is opposite the right angle. Getting these names precise and angle-relative now is what makes sine, cosine and tangent make sense in the lessons ahead.

Names that depend on the angle

The heart of this lesson is a subtle shift: a side's name is not a fixed label stuck to it, but a description of its **relationship to a chosen angle**. It is worth slowing down on the difference between the two kinds of name. The **hypotenuse** is **absolute** — it is opposite the right angle, full stop, whichever acute angle you consider. The **opposite** and **adjacent** are **relative** — they only make sense once you have chosen an angle θ, and they exchange roles if you choose the other one. Letting the learner see that swap for themselves (name one triangle from angle A, then from angle B) makes the idea land far better than a stated rule. The second thread is orientation: by naming sides in rotated and reflected triangles, the learner is forced to use the definitions rather than screen position. If a learner is hesitant, anchor every judgement to two questions — 'where is the right angle?' (that fixes the hypotenuse) and 'where is θ?' (across from it is the opposite, next to it is the adjacent).

The shape of the lesson

Timings are invitations, not deadlines. A learner may need much longer on one phase and skim another — that is expected. Each Cornerstone is given an honest mathematical role here:

Cornerstone	Its role here	Invitational timing
Connection	Side Labeller: watch the three sides named from θ; switch the angle and change orientation to see what stays and what swaps.	~12 min
Movement	Label it yourself: name a highlighted side as hypotenuse, opposite or adjacent, in triangles rotated every way.	~12 min
Reflection	Which side is it?: judge a claim about a side as right or a slip, catching the adjacent-for-hypotenuse error.	~10 min
Creativity	Same triangle, two viewpoints: name one triangle from each acute angle and see opposite and adjacent swap.	~6 min
Rest	A pause after realising a side can have two different names depending on where you look from.	~3 min
Nutrition	Mode B (intellectual): what the learning feeds — precision of language and naming by role, not by position.	~2 min

Common misconceptions, and gentle repairs

These are the sticking points to expect. In every case, the aim is a question that returns the thinking to the learner — never to supply the answer.

Naming sides by position ('the bottom one', 'the tall one')

Repair: “If we rotate the triangle, does that name still fit? Name it by its job — opposite the right angle, or across from θ, or next to θ.”

Confusing the adjacent with the hypotenuse (both touch θ)

Repair: “Which side is opposite the right angle? That one is the hypotenuse. The other side touching θ is the adjacent.”

Thinking the opposite and adjacent are fixed to particular sides

Repair: “Choose the other acute angle. Which side is now across from it? The names swap.”

Thinking the hypotenuse changes when you switch angle

Repair: “Is it still opposite the right angle? Then it is still the hypotenuse — that name never moves.”

Assuming the opposite must be vertical (or the adjacent horizontal)

Repair: “The words vertical and horizontal describe the page; opposite and adjacent describe the angle. Rotate it and check.”

Protecting the support pathway

The lesson offers support in a deliberate order: **Socratic prompt** → **first try** → **hint** → **second try** → **worked solution**. The worked solution for each question stays locked until the learner has used the prompt, used the hint, and checked two answers. This is by design: the steps *are* the learning.

Please hold this line

Do not read the worked solution aloud, or talk a learner straight to the answer, while it is still locked. If they are stuck, use the Reconnection Routes, the prompt and the hint, or ask one of the repair questions above. Arriving at a correct line of reasoning — even slowly — is worth far more than being handed the result.

Reconnection Routes

If an earlier idea is blocking progress, the lesson’s Reconnection Routes offer short recaps of exactly the prerequisites this lesson needs: right-angled triangles, the right angle (90°) and its square marker, the hypotenuse as the longest side, and reading a marked angle θ (from Lesson 1 and the Pythagoras unit). A learner can choose one independently, or you can invite one without requiring it. Using a route is part of learning, not evidence of falling behind, and it always returns the learner to their place.

Supporting through questions: an example

Anchor every naming decision to the right angle and to θ:

- “Where is the right angle? So which side is the hypotenuse?”
- “Where is θ? Which side is across from it — the opposite?”
- “Which side is next to θ but is not the hypotenuse — the adjacent?”
- “Now look from the other angle. Which two names swap, and which stays?”

If they stall, that is the moment for a Reconnection Route or the lesson’s prompt and hint — not for the answer.

Questions you might have

“Why do two of the names change but the hypotenuse does not?”
Because the hypotenuse is defined by the right angle, which never moves. The opposite and adjacent are defined by the acute angle you choose, so choosing the other one swaps them.

“How can the same side have two different names?”
A side's name describes its relationship to the angle you are looking from, not the side itself. From one angle it is the opposite; from the other it is the adjacent — both are correct, for their own angle.

“Why draw the triangle rotated and reflected?”
So names are learned by their role, not their position. A learner who can label a triangle in any orientation truly understands the definitions, which is what the next lessons rely on.

Keeping things regulated and calm

- Keep to one clear step at a time; there is no time pressure and no benefit to speed.
- Treat a wrong or unfinished answer as information for the next step, never as failure.
- Let the learner choose how to record — typed, written, drawn, spoken or annotated.
- If frustration rises, it is fine to pause and return later; the lesson holds their place.
- Avoid any language about age, school year or pace as a judgement of ability.

Recognising progress

Progress is described as Emerging → Developing → Secure → Mastering across understanding, application, communication and reflection. These describe where a learner is right now, not labels of what they are. Real understanding shown in part is still real understanding, even while other pieces are still settling.

Where they are	What it can look like
Emerging	Beginning to engage; names some features and makes a start with support.

Developing	Carries out the method with prompts; can explain part of the reasoning.
Secure	Works through independently and explains why each step is true.
Mastering	Applies the idea confidently to a new or unfamiliar case, and could teach it to someone else.

A note on evidence

Evidence can be captured in whatever form suits the learner — a photo of working, a short spoken explanation, an annotated Scratchpad image, or a written solution. There is no single required format. Incomplete work is information for the next connection, never a failure: it tells you exactly where to begin next time. The aim is a true record of the learner's thinking, not a tidy performance.

If a learner is stuck or distressed

Slow the pace and reduce the load: return to one concrete, familiar case and a single small question. Offer a Reconnection Route. Remind them they can pause and come back. A short, successful, low-stakes step rebuilds confidence more reliably than pushing through. Connection before curriculum — always.

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